```r
gb <- filter(clean_transit, clean_transit$country == "UK") %>%
  mutate(country = "GB")

clean_transit<- subset(clean_transit, !e %in% c(7226, 7336, 7417))
clean_transit <- rbind(clean_transit, gb)


clean_transit$region <- countrycode(sourcevar = clean_transit$country,
                            origin = "iso2c",
                            destination = "region23")
```

```r
clean_transit <- clean_transit %>%
  mutate(group = case_when(
    region %in% c("Northern America", "Central America", "South America") ~ "Americas",
    region %in% c("Southern Europe", "Northern Europe", "Western Europe") ~ "Europe",
    region %in% c("Central Asia", "Eastern Asia") ~ "Asia",
    region %in% c("South-Eastern Asia", "Australia and New Zealand") ~ "SE Asia & Oceania",
    region %in% c("Northern Africa", "Western Asia") ~ "North Africa & West Asia",
    .default = region
  ))



#table(clean_transit$group)
df <- clean_transit
df <- df[!is.na(df$tunnel_per), ]
```

Abstract

Write a concise summary of your project here (usually 150-250 words). It should state the research question, briefly describe the data, mention the primary linear regression models used, and summarize the key findings and conclusions.

Introduction

Data

Our transit data comes from the Transit Costs Project collaboration at the NYU Marron Institute of Urban Management. The express purpose of this dataset is to document the costs of transit projects across the world for international comparison. The transit projects under consideration all involve a railway, including light rail, trams, rapid transit/metro systems, and heavy rail (the meaning of this terminology changes based on local vernacular). This means busses, ferries, rideshare, bikes or other forms of transportation are not included in the dataset. This data was compiled by academic researchers and documents transit projects from . Each observation is a unique transit project undertaken by that country

To account for spatial factors, country shapefiles from a public ArcGIS dataset were merged into the . The map projection used in the dataset is the Mercator projection, which emphasizes accuracy of shape over size. The final dataset that was merged with the transit dataset was a cross-cultural psychology dataset from social psychologist Geert Hofstede. This dataset includes Hofstede's rankings of countries along six cultural dimensions: individualism-collectivism, uncertainty avoidance, strength of social hierarchy (called 'power-distance'), motivation toward achievement and success (formerly masculine v. feminine), and long-term vs. short term orientation. The observations in the Hofstede dataset were individual countries (or groupings of countries such as "Arab states").

Exploratory Analysis## Data Cleaning

We remove a number of variables such as `source` and `currency` that serve only to tell us the source of the data or the local currency, so we can work with variables relevant to our questions such as `real_cost` and `length`. We then set the `real_cost` variable to be a number, and filter out observations with missing variables or that have yet to be completed (as we do not care for projects yet to begin or yet to finish).

We create three new variables: `duration` (dbl): how long the project took, computed as the difference between the end and start years `region` (character): which geographic region the transit project's country is located in, as per the country's ISO code `group` (character): a further grouping of projects by geographic regions, to ensure that there are sufficient observations to represent different geographic areas

Introduction

Project Goal

To analyze and compare transit projects around the world to understand the factors that affect the cost and speeds of projects around the world, including the additional impact of geographic or cultural factors.

Research Questions

1. Project Speed What variables are significant when predicting the speed at which transit projects are completed?

2. Project Cost Which variables are significant when predicting the cost of a transit project?

3. Spatial and Cultural factors Are spatial and cultural factors significant for models that predict the cost and speed of a project?

Methodology

Three models were fit to the combined data. Each of these models tackle each of our connected research questions.

All of our models encode

Model 1 - Project Speed

$$Y_i = \beta_0 + \beta_1 \cdot X_{1i} + \beta_2 \cdot X_{2i} + \beta_3 \cdot X_{3i} + \sum_{j=4}^{9} \beta_j \cdot X_{ji}$$

Where:
X_{1i} = the natural log of the real cost of the project in millions of USD X_{2i} = length of the project in km X_{3i} = indicator of whether the project is based on railroad or not (0 if not, 1 if yes) And the sum notes the effect of the regional group the project it belongs to. There are 7 regional groups, with the base case being the Americas

Model 2 - Cost Prediction

$$Y_i^{\frac{1}{4}} = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \beta_6 X_{6i} + \beta_7 X_{7i} + \beta_8 X_{8i} + \beta_9 X_{9i} + \beta_{10} X_{10i} + \beta_{11} X_{11i}$$

Where: Y_i = The real cost of a project. Real cost is the nominal cost adjusted based on inflation indexes between countries. $X_{1i}, X_{2i}, \ldots, X_{6i}$ are dummy variables for the region group the country belongs in. $X_{7i} = $ Midpoint year of construction for the project $X_{8i} = $ Length of transit route built (km) $X_{9i} = $ Number of stations built X_{10i} = Purchasing Power Parity (PPP), based on midpoint of construction X_{11i} = Duration of transit project (years)

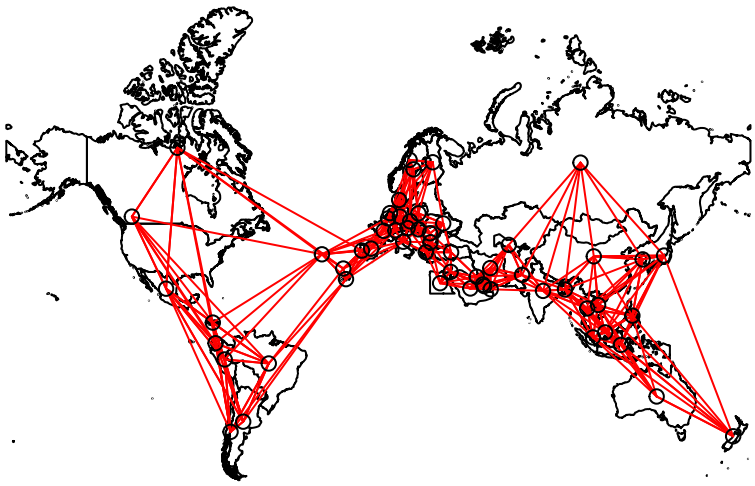
To address non-constant variance and improve residual normality, a Box–Cox transformation was applied to the response variable. This subset of covariates was chosen through backward stepwise regression using AIC as the selection criteria. Cook's distance was used to identify outliers and VIF was used to assess multicollinearity but neither outliers or multicollinearity were identified in this model. After implementing the Box-Cox transformation,

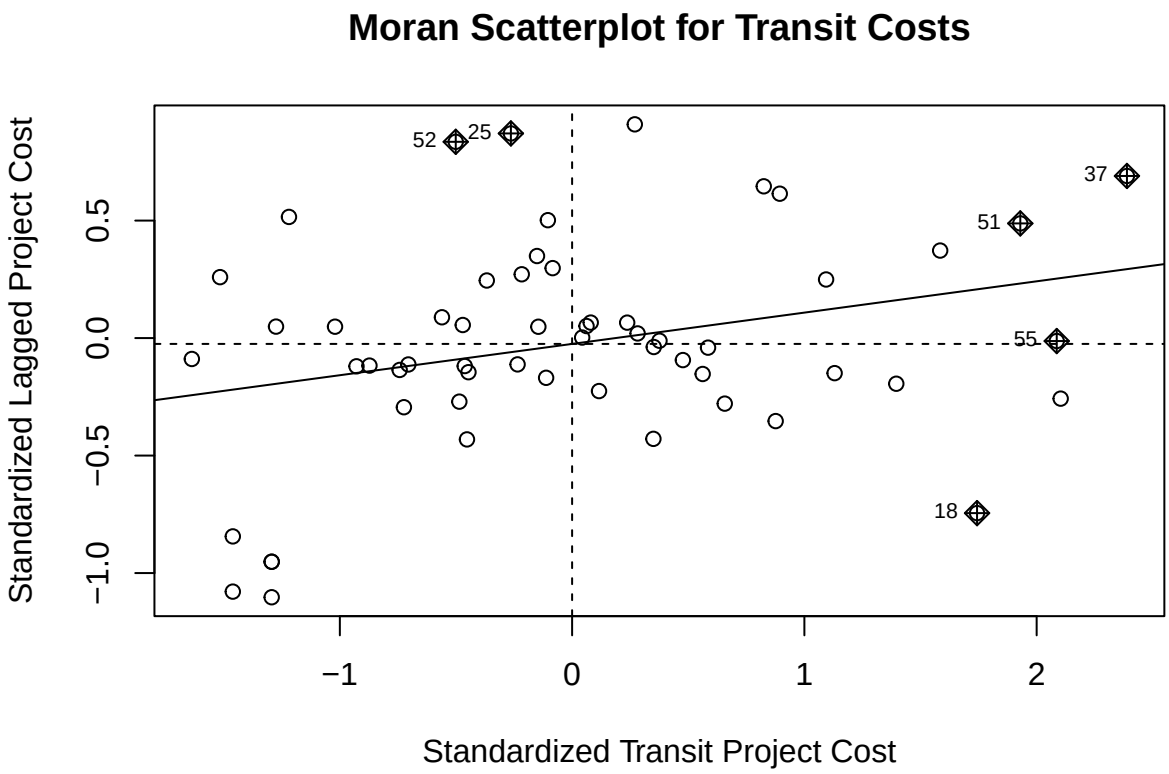
Accounting for spatial

A particular concern for our dataset is spatial autocorrelation. There is strong reason to believe that the presence of transit in neighboring countries could disincentivise or incentivise countries to build it themselves. If a country's passenger rail buildout is successful, it encourages neighbors to invest in rail projects themselves. Certain countries have legal frameworks or cultural biases that discourage infrastructure investments like transit, and these features are often shared among neighboring countries. Spatial autocorrelation may also capture the aspects of geography that make it more costly or inexpensive to build in certain regions, such as soil quality or climate.

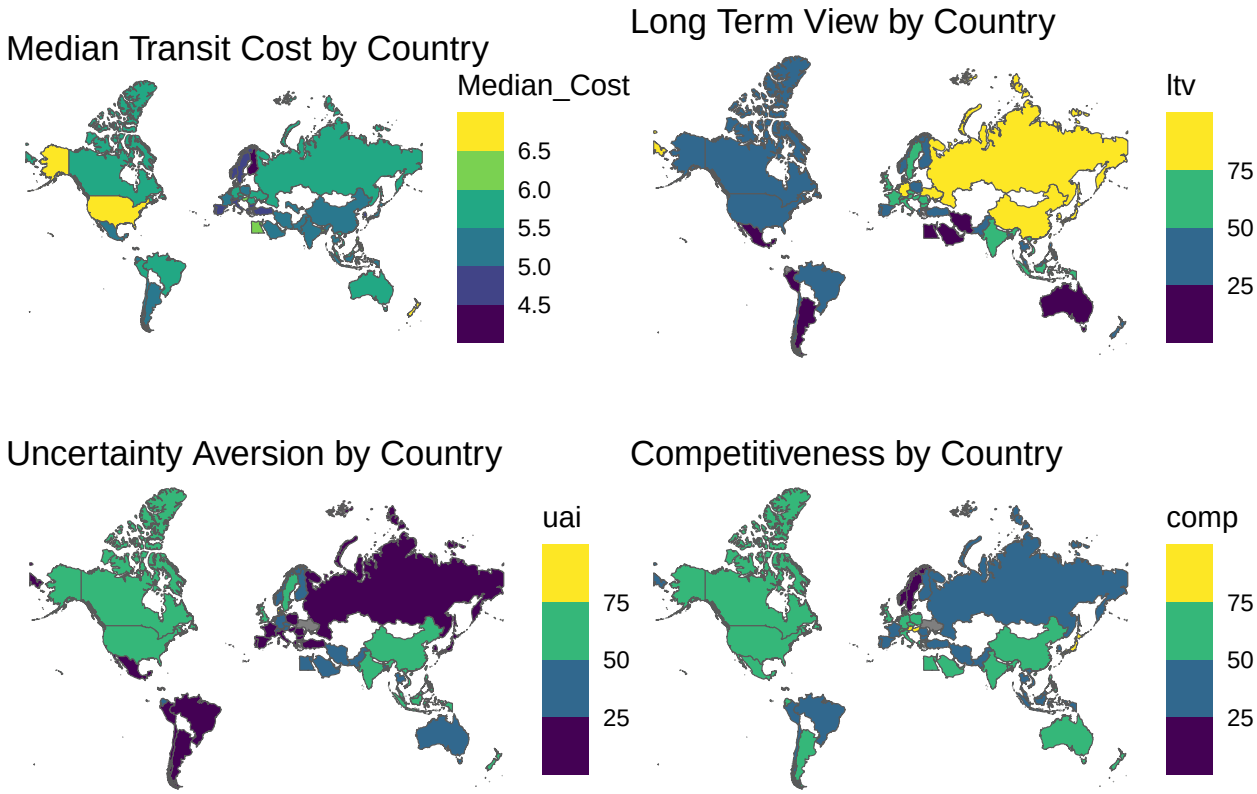
To model spatial autocorrelation, we use the shapefiles to

```r
plot(st_geometry(medshp))
plot(dist_nb, coords = st_coordinates(centroid), add = TRUE, col = "red")
```






```
## Warning: There were 5 warnings in ‘mutate()‘.
## The first warning was:
## i In argument: ‘longtermview = as.numeric(ltowvs)‘.
## Caused by warning:
## ! NAs introduced by coercion
## i Run ‘dplyr::last_dplyr_warnings()‘ to see the 4 remaining warnings.
```



Some econometric procedures recommending accounting for spatial autocorrelation by inflating standard errors.

```
##                     GVIF Df GVIF^(1/(2*Df))
## uncertainty     2.379687  1        1.542623
## power_distance  2.232496  1        1.494154
## competitive     1.535409  1        1.239116
## year            1.226286  1        1.107378
## length          1.305438  1        1.142558
## ppp_rate        2.862822  1        1.691987
## rr              1.225111  1        1.106847
## tunnel_per      1.465959  1        1.210768
## duration        1.466593  1        1.211030
## group          14.763651  6        1.251506
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 110.3872, Df = 1, p = < 2.22e-16
```

Model 3 -

Model 4 -

Results

Project Speed (Sangey Rinchen)

With this model, we are interested in answering the question of whether or not a model can predict how quickly a transit project can be completed, and which predictors are required to answer that question. There isn't a specific variable for that in the dataset, so we begin by creating the variables we need and begin creating our model.

Data cleaning and the full model We need a variable for the project speed, which we calculate with the simple formula:

$$\frac{\text{length}}{\text{duration}}$$

:

```
df_speed <- df %>%
  filter(!is.na(duration)) %>%
  mutate(project_speed = length / duration)

rownames(df_speed) <- NULL #resetting row names
```

We begin with our initial model, factoring in possible explanatory variables (a few such as year and duration are excluded, and attempt to find the best subset using BIC, adjusted R squared, and Mallow's C_p. Note that we use the natural log of the real cost to avoid any possible issues that may arise due to how much bigger the numbers in this variable are compared to the rest of the dataset.

```
subsets <- summary(regsubsets(project_speed ~ log(real_cost) + rr + length +
                      stations + cost_km_millions + group, nbest = 1, nvmax = 8, data = df_speed))
subsets$adjr2
```

```
## [1] 0.6454171 0.6653464 0.6754917 0.6796836 0.6833430 0.6870058 0.6880574
## [8] 0.6882169
```

```
subsets$cp
```

```
## [1] 56.692822 31.672830 19.457158 15.001127 11.258294  7.534701  7.184215
## [8]  7.982220
```

```
subsets$bic
```

```
## [1] -413.0328 -431.6843 -439.2703 -439.5854 -439.2848 -439.0457 -435.4272
## [8] -430.6438
```

```r
subsets$outmat
```

```
##          log(real_cost) rr1 length stations cost_km_millions groupAsia
## 1  ( 1 ) " "            " " "*"    " "      " "               " "
## 2  ( 1 ) " "            " " "*"    " "      " "               "*"
## 3  ( 1 ) " "            " " "*"    "*"      " "               "*"
## 4  ( 1 ) " "            "*" "*"    "*"      " "               "*"
## 5  ( 1 ) "*"            "*" "*"    "*"      " "               "*"
## 6  ( 1 ) "*"            "*" "*"    "*"      " "               "*"
## 7  ( 1 ) "*"            "*" "*"    "*"      " "               "*"
## 8  ( 1 ) "*"            "*" "*"    "*"      "*"               "*"
##          groupEastern Europe groupEurope groupNorth Africa & West Asia
## 1  ( 1 ) " "                 " "         " "
## 2  ( 1 ) " "                 " "         " "
## 3  ( 1 ) " "                 " "         " "
## 4  ( 1 ) " "                 " "         " "
## 5  ( 1 ) " "                 " "         " "
## 6  ( 1 ) " "                 " "         "*"
## 7  ( 1 ) " "                 "*"         "*"
## 8  ( 1 ) " "                 "*"         "*"
##          groupSE Asia & Oceania groupSouthern Asia
## 1  ( 1 ) " "                    " "
## 2  ( 1 ) " "                    " "
## 3  ( 1 ) " "                    " "
## 4  ( 1 ) " "                    " "
## 5  ( 1 ) " "                    " "
## 6  ( 1 ) " "                    " "
## 7  ( 1 ) " "                    " "
## 8  ( 1 ) " "                    " "
```

Though there isn't one lone model all metrics agree on, subset 6 has satisfactory scores and will be used (though with all groups in the `group` variable present).

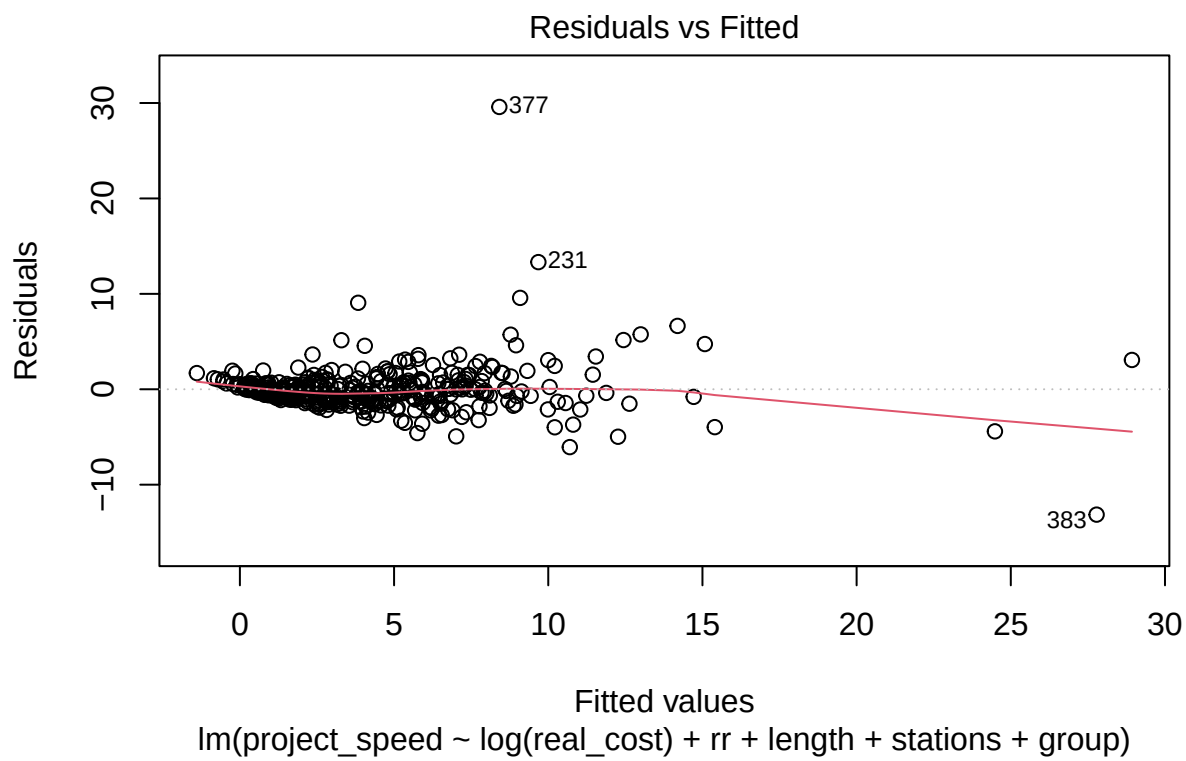
```r
speedMdl <- lm(project_speed ~ log(real_cost) + rr + length + stations + group, data = df_speed)
summary(speedMdl)
```

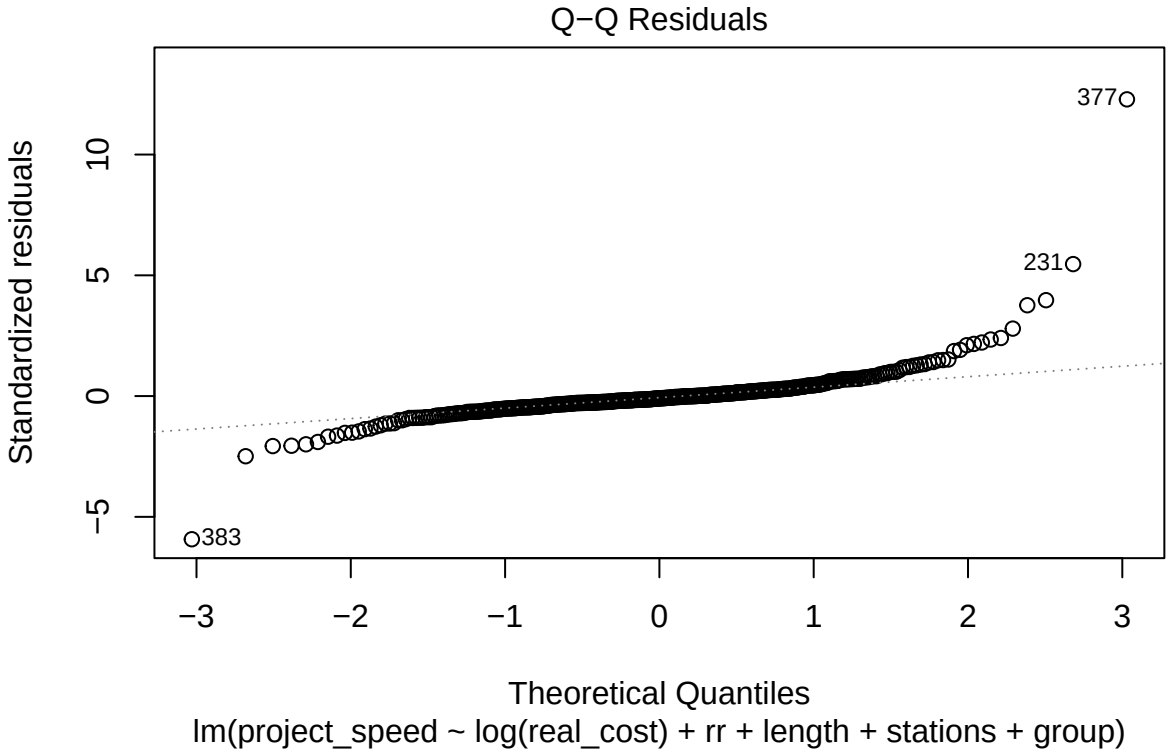
Building the model

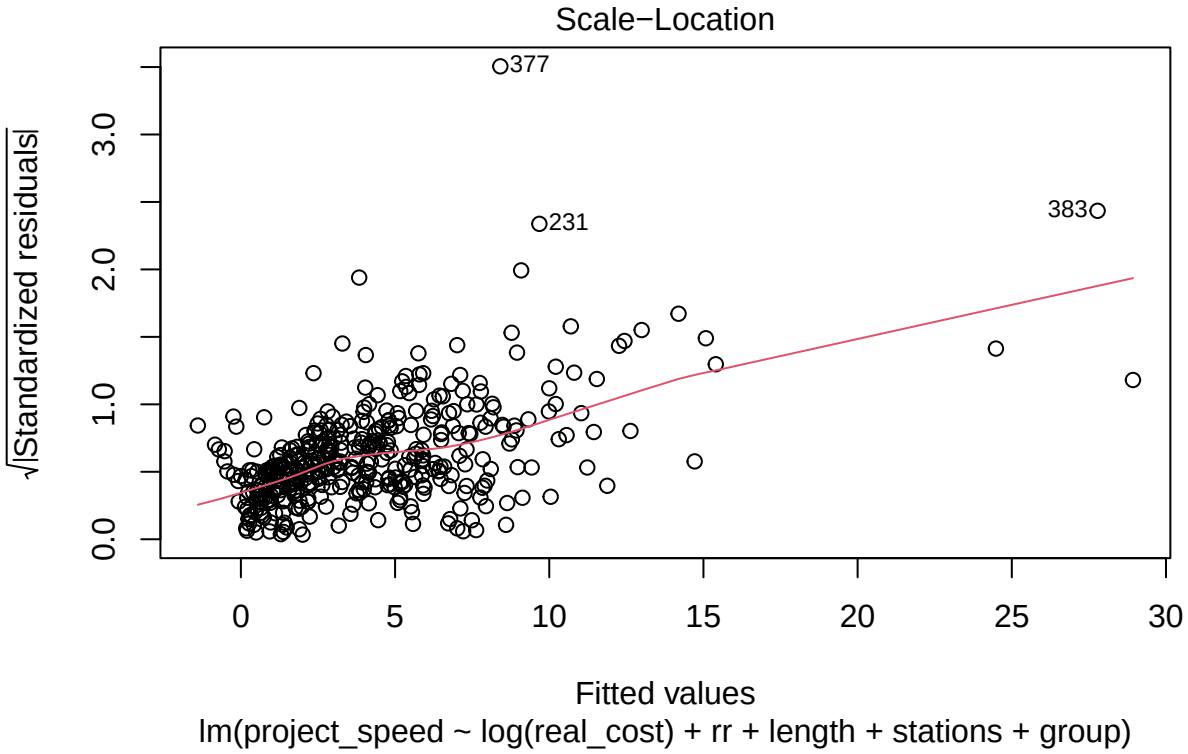
```
##
## Call:
## lm(formula = project_speed ~ log(real_cost) + rr + length + stations +
##     group, data = df_speed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.1367  -0.8670  -0.2308   0.5517  29.5864
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  -3.88221    1.35510  -2.865  0.00439 **
```

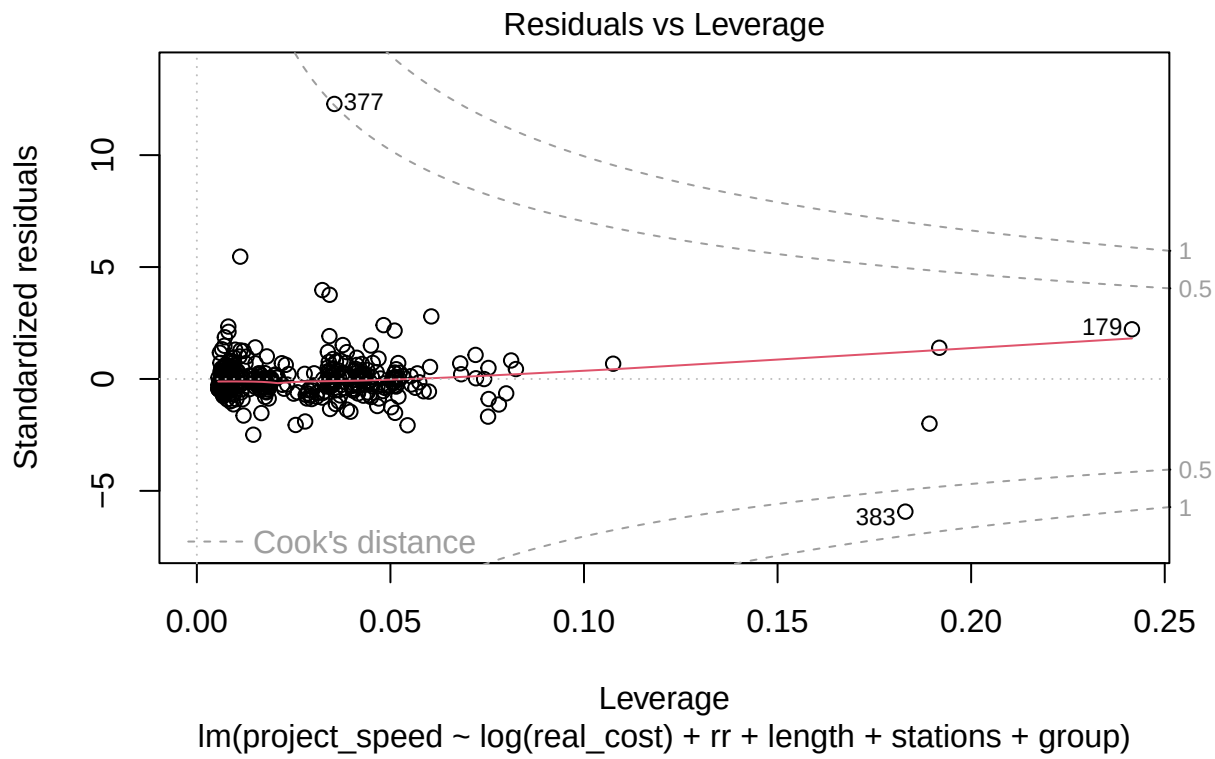
```
## log(real_cost)                 0.53956    0.16903   3.192  0.00152 **
## rr1                           -1.40644    0.49105  -2.864  0.00440 **
## length                         0.19459    0.01294  15.038  < 2e-16 ***
## stations                      -0.08107    0.01862  -4.353 1.71e-05 ***
## groupAsia                      1.62148    0.49075   3.304  0.00104 **
## groupEastern Europe            0.70364    0.72247   0.974  0.33068
## groupEurope                    0.97899    0.58330   1.678  0.09406 .
## groupNorth Africa & West Asia  1.61956    0.62412   2.595  0.00981 **
## groupSE Asia & Oceania        -0.22659    0.66897  -0.339  0.73500
## groupSouthern Asia             0.59445    0.65947   0.901  0.36792
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.452 on 398 degrees of freedom
##   (4 observations deleted due to missingness)
## Multiple R-squared:  0.6952, Adjusted R-squared:  0.6875
## F-statistic: 90.78 on 10 and 398 DF,  p-value: < 2.2e-16
```

```r
plot(speedMdl)
```




Q−Q Residuals
Standardized residuals
10
5
0
−5
377
231
383
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(project_speed ~ log(real_cost) + rr + length + stations + group)


Scale−Location
377
231
383
√|Standardized residuals|
0.0
1.0
2.0
3.0
0
5
10
15
20
25
30
Fitted values
lm(project_speed ~ log(real_cost) + rr + length + stations + group)




```r
#test for multicollinearity
vif(speedMdl)
```

```
##                    GVIF Df GVIF^(1/(2*Df))
## log(real_cost) 2.630298  1        1.621819
## rr             1.080618  1        1.039528
## length         4.745510  1        2.178419
## stations       4.590693  1        2.142590
## group          1.845534  6        1.052390
```

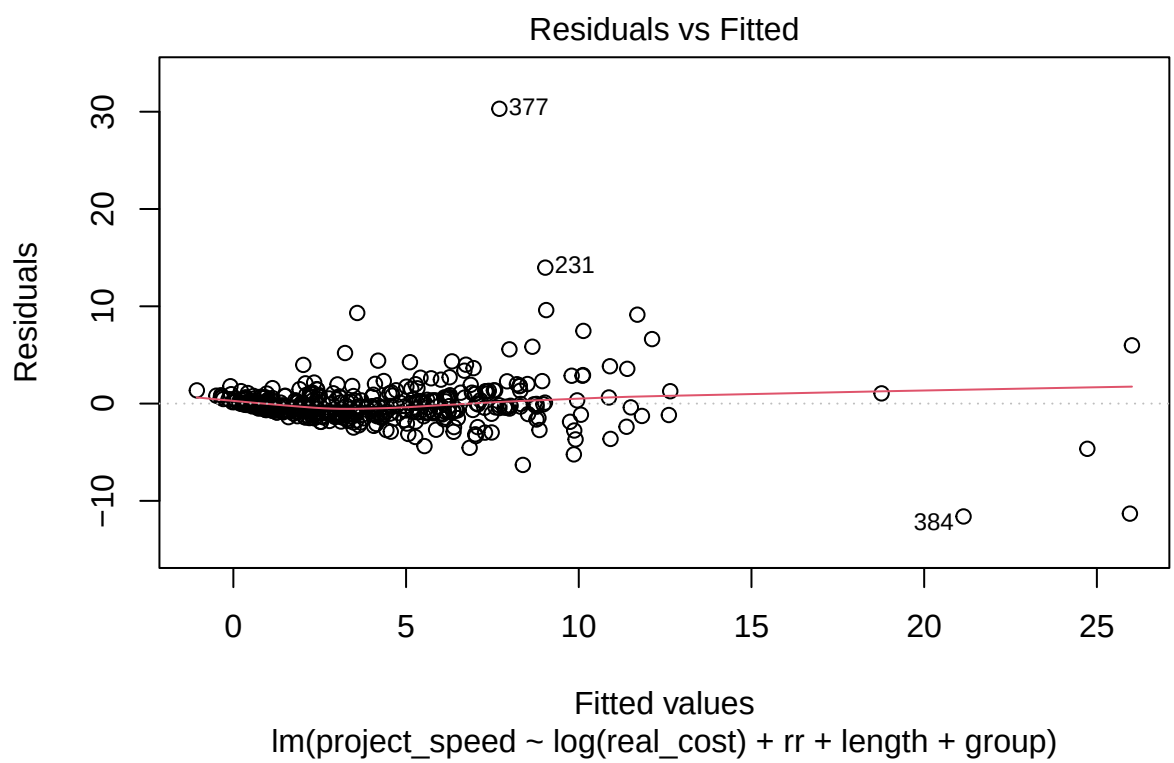
Apart from some potential outliers and a bit of skewing on the Q-Q plots, there isn't anything too concerning. However, we observe some collinearity between `length` and `stations`, understandable given that the two can be used to understand the size of the project in their own ways. Thus, we will remove the `stations` variable and diagnose our model.

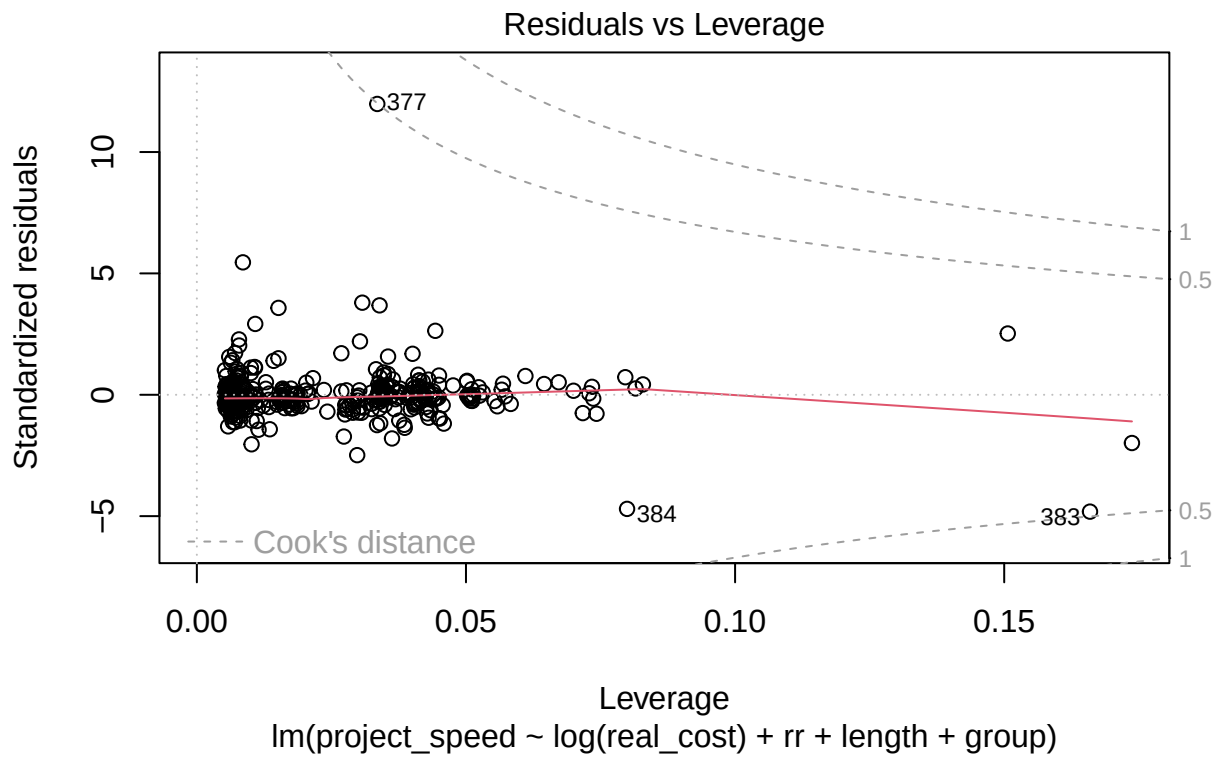
```r
speedMdl2 <- lm(project_speed ~ log(real_cost) + rr + length + group, data = df_speed)
summary(speedMdl2)
```

```
##
## Call:
## lm(formula = project_speed ~ log(real_cost) + rr + length + group,
##     data = df_speed)
##
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -11.6013  -0.8878  -0.2442   0.5698  30.2994
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -2.899238   1.408316  -2.059 0.040169 *
## log(real_cost)                 0.388548   0.174260   2.230 0.026318 *
## rr1                           -1.093162   0.500737  -2.183 0.029604 *
## length                         0.147144   0.008918  16.499  < 2e-16 ***
## groupAsia                      1.783204   0.513543   3.472 0.000572 ***
## groupEastern Europe            0.743316   0.757643   0.981 0.327137
## groupEurope                    0.768250   0.610548   1.258 0.209013
## groupNorth Africa & West Asia  1.360733   0.653149   2.083 0.037850 *
## groupSE Asia & Oceania        -0.124132   0.701602  -0.177 0.859655
## groupSouthern Asia             0.133148   0.676415   0.197 0.844049
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.573 on 403 degrees of freedom
## Multiple R-squared:  0.6668, Adjusted R-squared:  0.6594
## F-statistic: 89.63 on 9 and 403 DF,  p-value: < 2.2e-16
```

```r
plot(speedMdl2, which = c(1, 5))
```






```r
vif(speedMdl2)
```

```
##                    GVIF Df GVIF^(1/(2*Df))
## log(real_cost) 2.610656  1        1.615753
## rr             1.054020  1        1.026655
## length         2.197769  1        1.482487
## group          1.641606  6        1.042171
```

Outliers Our updated model doesn't appear to have issues with multicollinearity. The residuals plot looks fine but we see some points with high leverage and high residuals. Let's take a closer look at these points to determine whether they should be removed as outliers.

```r
outlierTest(speedMdl2)
```

```
##      rstudent unadjusted p-value Bonferroni p
## 377 14.911996         2.4420e-40   1.0085e-37
## 231  5.661068         2.8659e-08   1.1836e-05
## 383 -4.952733         1.0795e-06   4.4584e-04
## 384 -4.829865         1.9458e-06   8.0363e-04
```

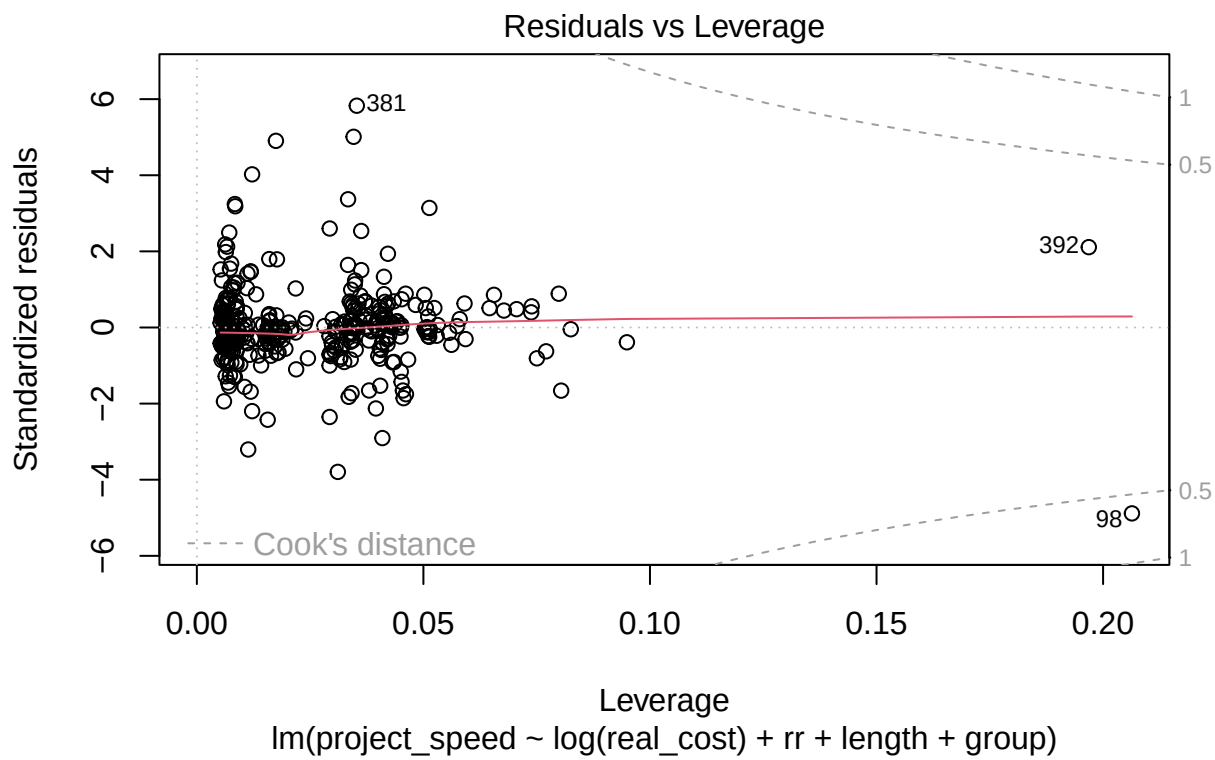
```r
df_speed[c(231, 377, 383, 384),]
```

```
##        e country     city start_year end_year rr  length tunnel_per  tunnel
```

```
## 231 7691      CN Shanghai       2007     2009  0  46.008      47.11  21.676
## 377 8171      SA   Riyadh       2015     2016  0  38.000      42.63  16.200
## 383 8186      SA   Jeddah       2014     2025  0 161.100      12.41  20.000
## 384 8193      QA     Doha       2013     2026  0 124.000      89.92 111.500
##     stations year ppp_rate real_cost cost_km_millions duration       region
## 231       21 2008   0.3143   5955.99         129.4600        2 Eastern Asia
## 377       22 2013   1.8200  11941.02         314.2374        1 Western Asia
## 383       81 2015   1.4350  17220.00         106.8901       11 Western Asia
## 384       NA 2013   2.5000  90000.00         725.8065       13 Western Asia
##                        group project_speed
## 231                     Asia     23.004000
## 377 North Africa & West Asia     38.000000
## 383 North Africa & West Asia     14.645455
## 384 North Africa & West Asia      9.538462
```

These observations have some of the fastest `project_speed`, longest `length`, and largest `real_cost` of any of the transit projects in the dataset. We will note these outliers, and remove them from our model for now.

```
speedMdl3 <- lm(project_speed ~ log(real_cost) + rr + length + group, data = df_speed[-c(231, 377, 383,
plot(speedMdl3, which = 5)
```

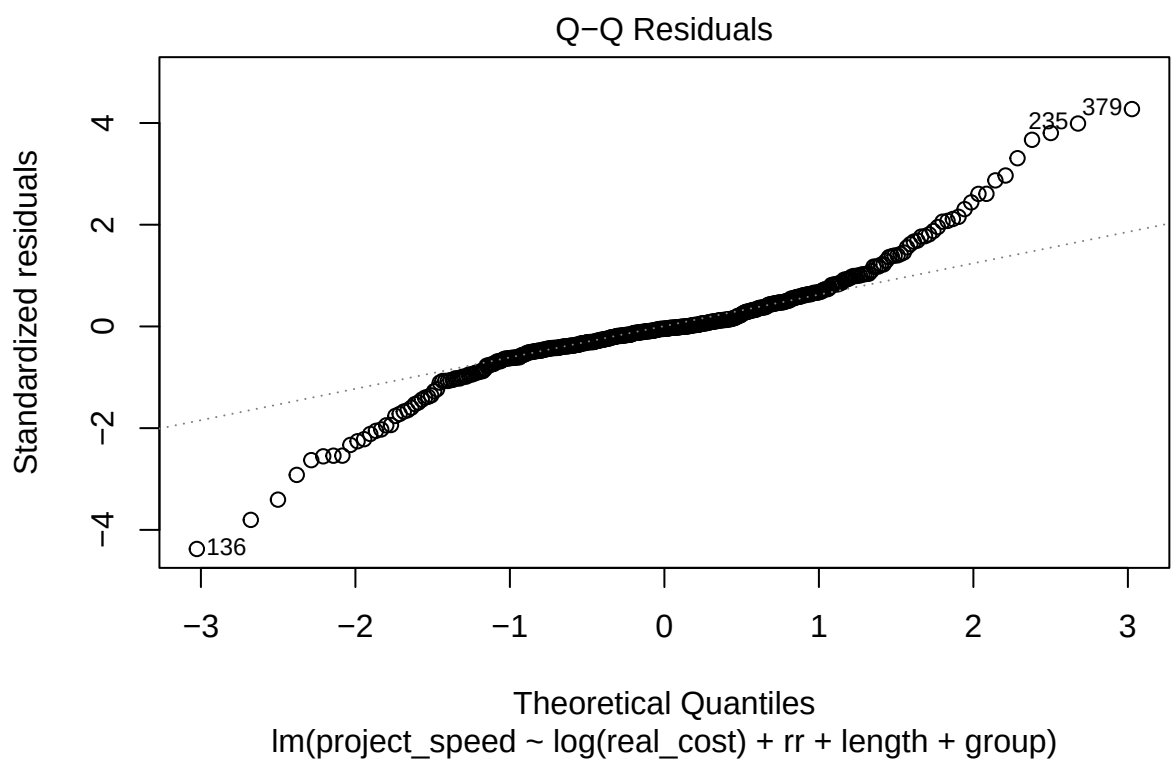


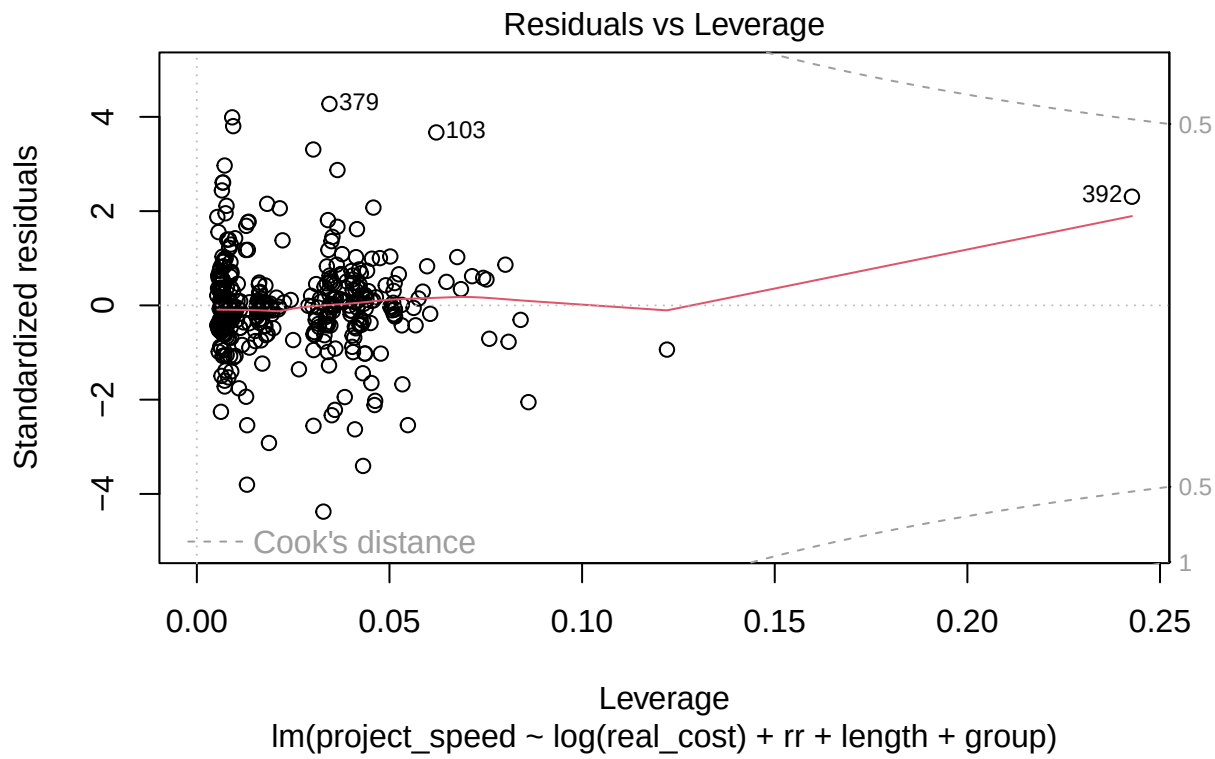

```
outlierTest(speedMdl3)
```

```
##      rstudent unadjusted p-value Bonferroni p
## 381  6.084494         2.7476e-09   1.1238e-06
```

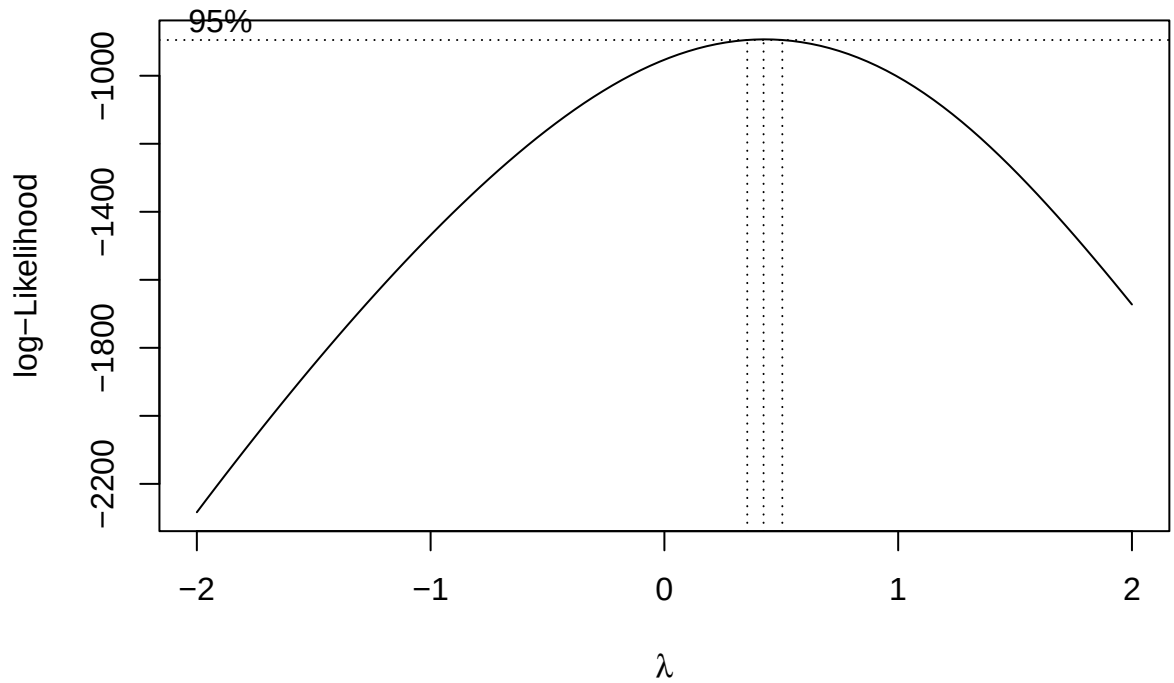
```
## 102  5.169453         3.7246e-07   1.5233e-04
## 303  5.052571         6.6560e-07   2.7223e-04
## 98  -5.031198         7.3928e-07   3.0237e-04
## 344  4.103210         4.9444e-05   2.0222e-02
```

```r
#We still see outliers, so we clear them out one more time

df_speed2 <- df_speed[-c(231, 377, 383,384),]

speedMdl4 <- lm(project_speed ~ log(real_cost) + rr + length + group, data = df_speed2[-c(98, 102, 302,
plot(speedMdl4, which = c(2,5))
```

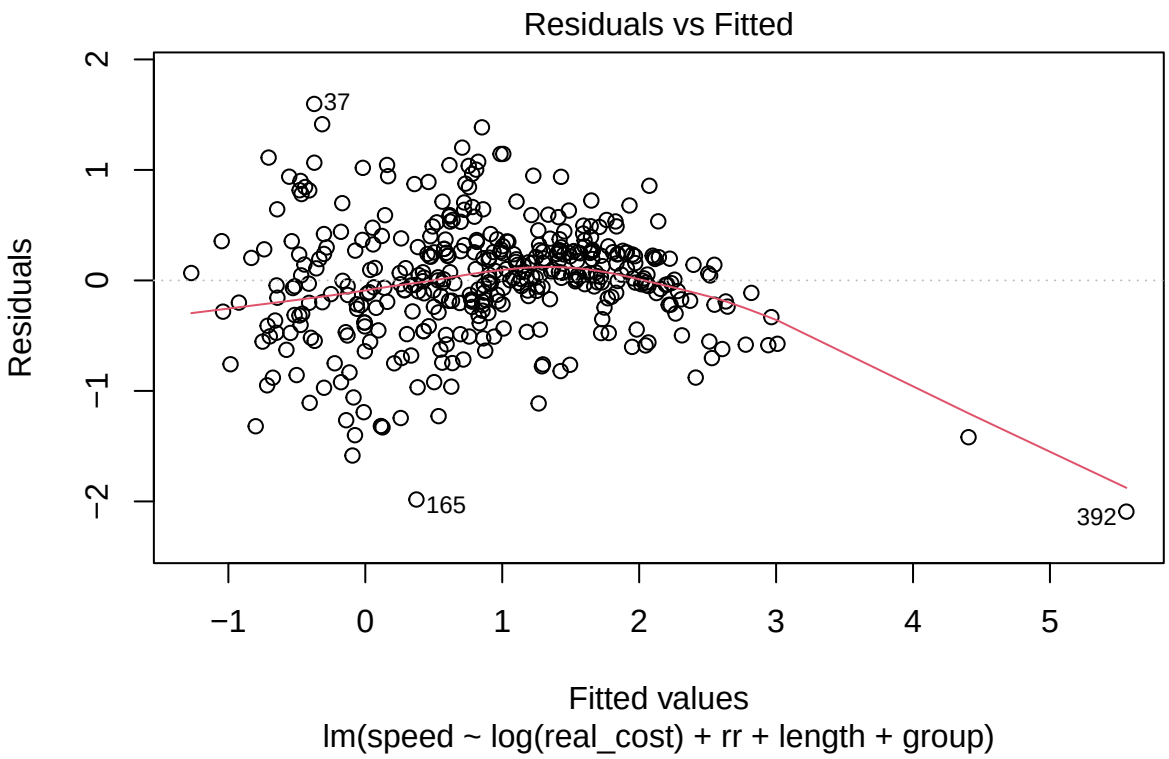




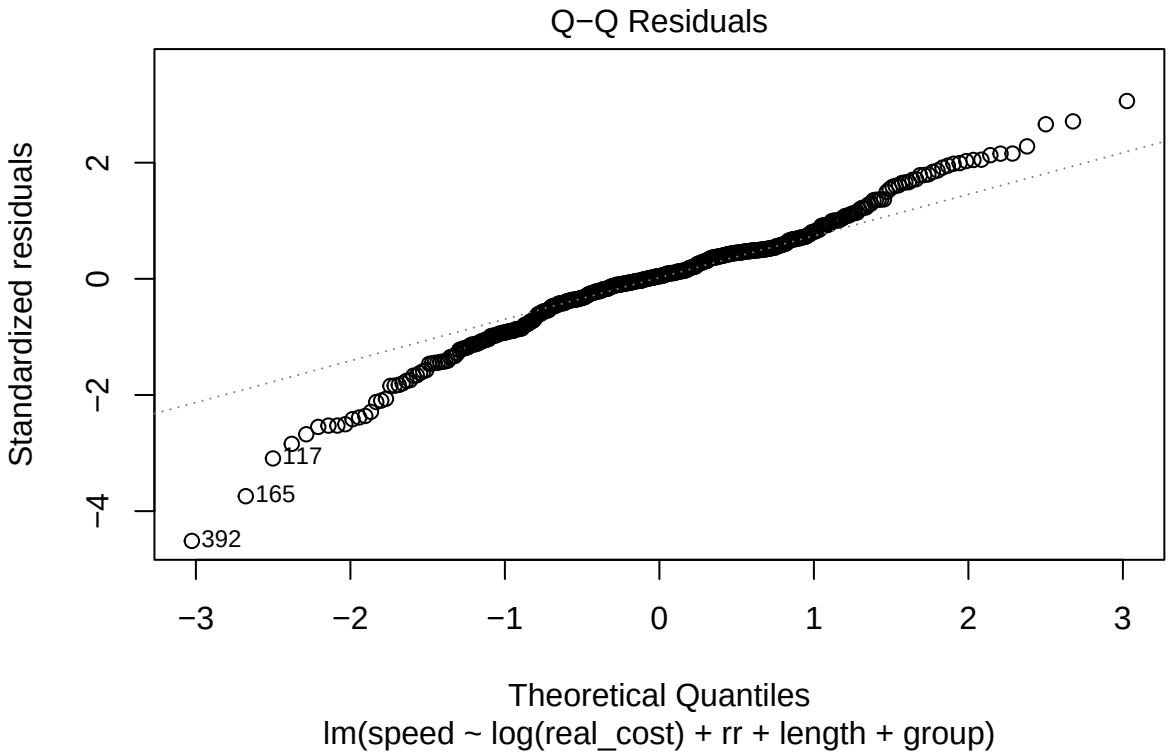

```
#Nothing glaring in the residuals vs leverage plot, so we're done with outliers. However, the Q-Q plot

boxcox(speedMdl4)
```

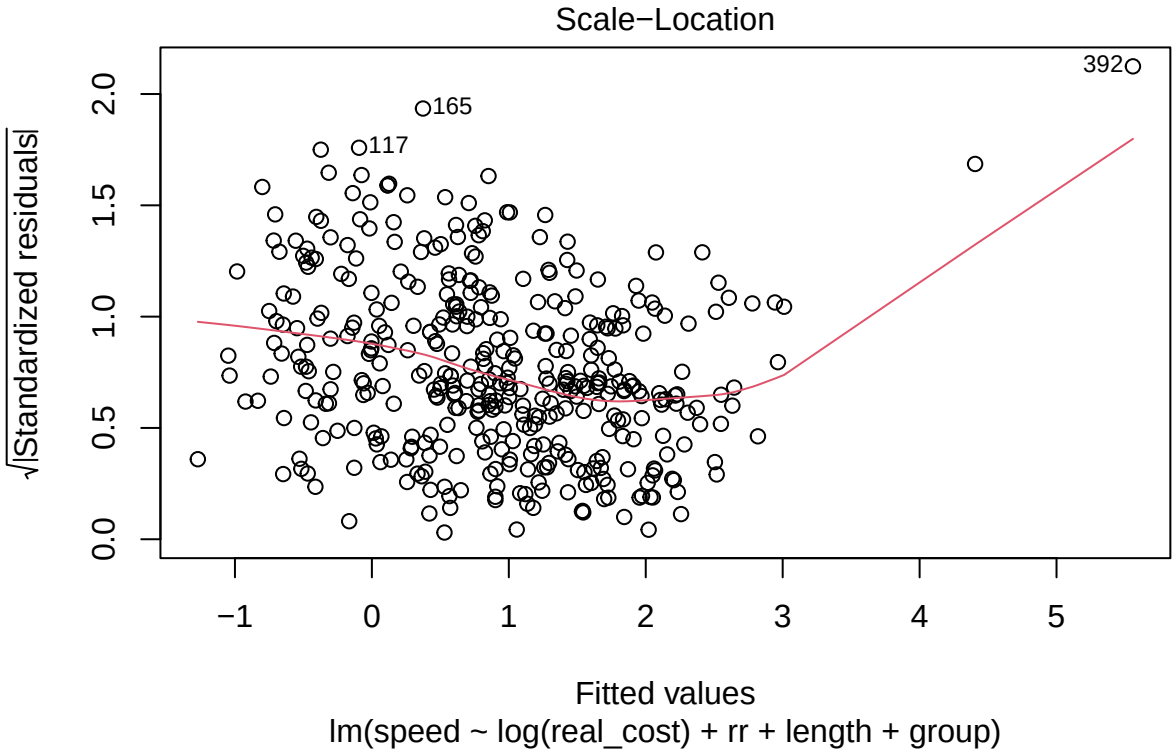


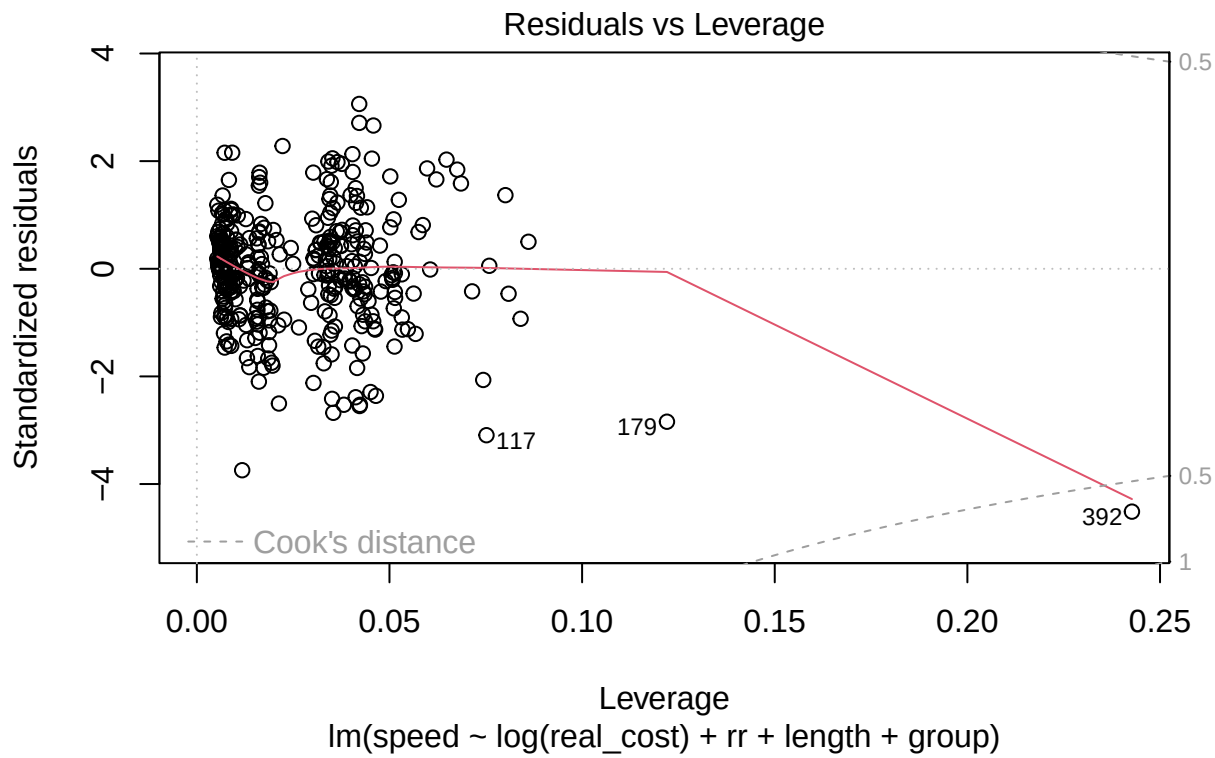

```
df_speed3 <- df_speed2 %>%
  mutate(speed = log(project_speed))
speedMdl_fin <- lm(speed ~ log(real_cost) + rr + length + group, data = df_speed3[-c(98, 102, 302, 379,

#Last checks:

plot(speedMdl_fin)
```


Residuals vs Fitted
Residuals
Fitted values
lm(speed ~ log(real_cost) + rr + length + group)
37
165
392

Final model


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(speed ~ log(real_cost) + rr + length + group)
117
165
392


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
392
165
117
−1
0
1
2
3
4
5
Fitted values
lm(speed ~ log(real_cost) + rr + length + group)




```r
vif(speedMdl_fin)
```

```
##                    GVIF Df GVIF^(1/(2*Df))
## log(real_cost) 2.786316  1        1.669226
## rr             1.056813  1        1.028014
## length         2.342033  1        1.530370
## group          1.640478  6        1.042112
```

```r
outlierTest(speedMdl_fin)
```

```
##      rstudent unadjusted p-value Bonferroni p
## 392 -4.627926         5.0241e-06    0.0020298
```

And so, our final model's coefficients are as follows:

```r
speedMdl_fin$coefficients
```

```
##                   (Intercept)                log(real_cost)
##                   -2.15315136                    0.26124395
##                           rr1                        length
##                   -0.44930744                    0.02777676
##                     groupAsia           groupEastern Europe
##                    0.81661074                    0.35824431
```

```
##                    groupEurope groupNorth Africa & West Asia
##                    -0.01875629                    0.57039119
##         groupSE Asia & Oceania            groupSouthern Asia
##                     0.32823305                    0.54173989
```

On observation, it makes sense that the project would take longer if the lines involved setting up railroads (hence `rr1` has a negative coefficient). We observe a typically faster project completion speed in projects in Asia , and a slower completion speed in Europe.

Cost Prediction ()

Fit the baseline linear model We model **real construction cost** as a function of (i) geographic regions and (ii) key characteristics that are plausibly associated with cost: project year, length of line, number of stations, purchasing power parity (PPP), and construction duration.

$$real_cost \sim region + year + length + stations + ppp_rate + duration$$

The initial model statistics is

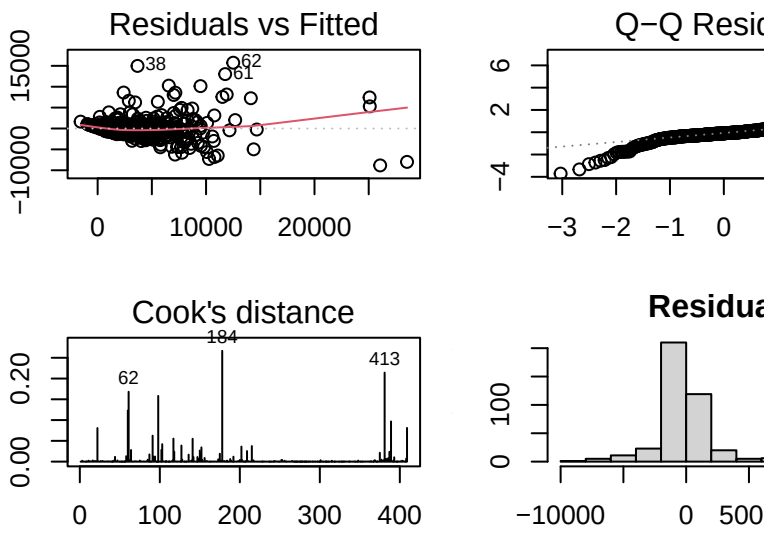
```r
full <- lm(real_cost ~ group + year+ length + stations+ ppp_rate+duration, data = df)

broom::glance(full) %>%
  dplyr::select(r.squared, sigma, AIC, BIC, nobs) %>%
  knitr::kable(digits = 3, caption = "Model fit statistics")
```

Table 1: Model fit statistics

r.squared	sigma	AIC	BIC	nobs
0.681	2592.439	7604.282	7656.46	409

Linear regression relies on zero means, constant error variance, and approximately normal residuals. We evaluate these using standard residual diagnostics. In the baseline model, residual plots indicate heteroskedasticity and departures from normality.

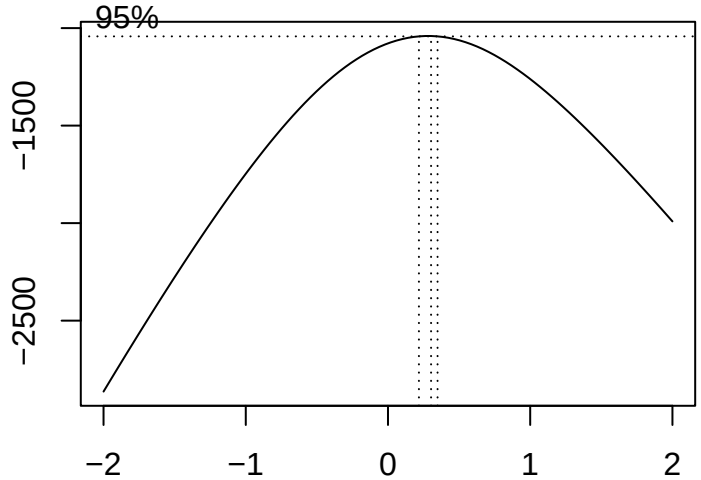
```r
par(mfrow = c(2,2), mar=c(3,3,2,1))
plot(full, which = 1)
plot(full, which = 2)
plot(full, which = 4)
hist(resid(full), main = "Residuals", xlab = "")
```



Diagnostic assessment of modeling assumptions

```
par(mfrow = c(1,1))
```

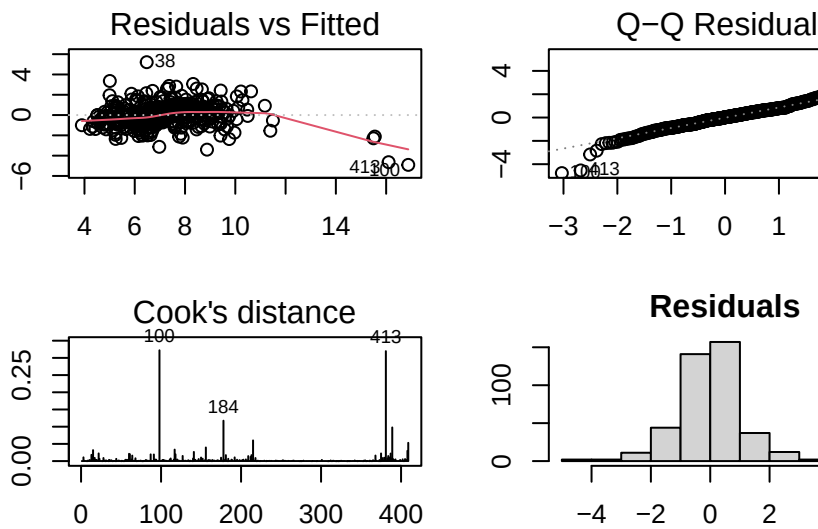
```
library(MASS)
df_pos <- df %>% filter(real_cost > 0)
full_pos <- lm(real_cost ~ group + year + length + stations + ppp_rate + duration, data = df_pos)
par(mar = c(3,3,2,1))
MASS::boxcox(full_pos, plotit = TRUE) ### The result turns out that we should raise y to power of 0.25
```



Transformation via BoxCox

The Box–Cox log-likelihood plot suggests a transformation near $\lambda \approx 0.25$, so we proceed with $Y^{0.25}$

```r
full_trans <- lm(real_cost^0.25 ~ group + year+ length + stations+ ppp_rate+duration, data = df)
#summary(full_trans)

par(mfrow = c(2,2), mar=c(3,3,2,1))
plot(full_trans, which = 1)
plot(full_trans, which = 2)
plot(full_trans, which = 4)
hist(resid(full_trans), main = "Residuals", xlab = "")
```



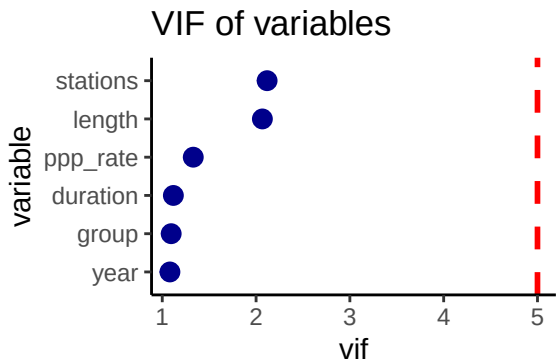
Refit our model and check conditions again.

```r
par(mfrow = c(1,1))
```

After transformation, the residual diagnostics improve noticeably. The TA plot suggests more constant variance, and the Q–Q plot shows residuals closer to normality.

We also examine influential observations. Although some points have larger standardized residuals, influence depends on both residual size and leverage, summarized by Cook's distance. All Cook's distances are well below common thresholds, 0.5, so no observations are removed.

```r
v <- vif(full_trans)

vif_df <- data.frame(
  variable = rownames(v),
  vif = v[,3]
)


ggplot(vif_df, aes(x = vif, y = reorder(variable, vif)))+
  geom_point(size = 3, color = 'darkblue')+
  geom_vline(xintercept = 5, linetype = 'dashed', color = 'red', linewidth=1 )+
  labs(title = 'VIF of variables', y ='variable')+
  theme_classic()
```



Multicollinearity assessment

We assess multicollinearity using variance inflation factors (VIF). Values exceeding 5 are often used as thresholds to indicate high collinearity. The VIF results here do not indicate problematic collinearity.

Model selection

Backward model selection To assess whether a simpler model provides similar explanatory power, we apply backward model selection using AIC starting from the transformed-response model. Because AIC balances model fit and complexity, smaller values indicate a better model.

```
step_fit <- step(full_trans,
     direction = 'backward', trace  = 0)

data.frame(
  Initial_AIC = round(AIC(full_trans),2),
  End_AIC = round(AIC(step_fit),2),
  Equation = paste(deparse(formula(step_fit)), collapse = ''))%>%
knitr::kable(caption = "Backward AIC selection summary")
```

Table 2: Backward AIC selection summary

Initial_AIC	End_AIC	Equation
1263.57	1263.57	real_cost^0.25 ~ group + year + length + stations + ppp_rate + duration

Selection result. The backward AIC procedure does not remove any predictors, indicating that each variable contributes meaningfully to model fit under the AIC criterion. It indicates that removing any single predictor increases AIC relative to the full transformed model.

Final Model

We proceed with the transformed-response model including all pre-specified predictors:

This model achieves improved adherence to the regression assumptions after the transformation, including approximately constant variance and residuals that are closer to normally distributed. Retaining all predictors also preserves interpretability with respect to the study objective, allowing us to evaluate how project characteristics and geographic region jointly influence transit construction costs. Below is the full summary of the model.

```r
# coefficient table
broom::tidy(full_trans) %>%
  dplyr::select(term, estimate, std.error, p.value) %>%
  knitr::kable(digits = 3,caption = "Summary of the transformed regression model.")
```

Table 3: Summary of the transformed regression model.

term	estimate	std.error	p.value
(Intercept)	-87.902	20.978	0.000
groupAsia	-0.503	0.239	0.036
groupEastern Europe	-0.897	0.328	0.007
groupEurope	-1.702	0.248	0.000
groupNorth Africa & West Asia	-0.816	0.294	0.006
groupSE Asia & Oceania	0.638	0.310	0.040
groupSouthern Asia	-0.813	0.315	0.010
year	0.047	0.010	0.000
length	0.049	0.006	0.000
stations	0.027	0.008	0.001
ppp_rate	-0.149	0.082	0.068
duration	0.069	0.020	0.001

```r
# model fit
broom::glance(full_trans) %>%
  dplyr::select(r.squared, adj.r.squared, sigma) %>%
  knitr::kable(digit = 3)
```

r.squared	adj.r.squared	sigma
0.694	0.686	1.115

Spatial and Cultural Predictors (Kevin Ji)

Conclusion